eateggsAI-30M v0.1
Experimental Low-VRAM GPT-Style LLM Whitepaper

Abstract
eateggsAI-30M is an experimental GPT-style Large Language Model (LLM) created as a low-VRAM AI challenge using consumer-grade hardware. The project explores whether transformer-based language models can be trained efficiently on older GPUs with limited VRAM. The model was trained from scratch using PyTorch on a GTX 1060 4GB GPU while maintaining stable transformer training and meaningful text generation capabilities.

1. Introduction
Modern AI systems are commonly trained using expensive high-performance GPUs and large-scale infrastructure. This project investigates the feasibility of training a GPT-style transformer model on older consumer hardware with limited memory capacity.

eateggsAI-30M was designed as an educational and experimental research project focused on: Understanding transformer architectures deeply Building a custom LLM pipeline from scratch Exploring low-VRAM AI optimization techniques Training transformer models on affordable hardware

2. Hardware Configuration

Component	Specification
Laptop / CPU	Acer i5
GPU	NVIDIA GTX 1060
VRAM	4GB

3. Model Architecture

Setting	Value
Parameters	30,044,544
Transformer Layers	6
Attention Heads	8
Embedding Size	384
Context Length	256
Dropout	0.1

4. Dataset Pipeline
The training dataset was created using multiple domains in order to expose the model to diverse language structures and topics.

Dataset Sources: Wikipedia AG News SQuAD PG19 Books Web Text **Pipeline Steps:** Dataset downloading and streaming Data cleaning and filtering Deduplication GPT2 tokenization Fixed-length block creation PyTorch tensor conversion

5. Training System
The model was trained using PyTorch with multiple optimization techniques for efficient low-VRAM training.

Training Features: Mixed Precision Training (AMP) Gradient Scaling AdamW Optimizer Cross Entropy Loss CUDA Acceleration Low-VRAM Optimizations

6. Experimental Results

Metric	Value
Training Time	~12 Hours
VRAM Usage	~2.7GB
Framework	PyTorch
Precision	Mixed Precision (AMP)
Final Loss	3.2 - 3.3

7. Sample Generations & Model Behavior

The generated outputs demonstrate that the transformer successfully learned sentence continuation, topic associations, scientific vocabulary patterns, and stylistic text generation behavior.

Observed behaviors include: Narrative continuation Scientific and encyclopedia-style formatting Historical and informational writing patterns Multi-style text generation Topic drifting common in small-scale base models Repetition and factual inconsistency limitations

Quantum physics generation example

```
Enter prompt: What is quantum physics?

Generated Text:

What is quantum physics? The process of the atom is the quantized in which a continuum of protons and bonding. The electron
can be used as a reductive for the atom, which converts to an electron pair, or an electron pair. If it has been coined by
an atom with electrons and has not been suggested by the Bohr model of quantum mechanics.  In addition to the
electromagnetic theory of a given element in which the atom is defined as a photon of proton, or a nucleus that has no
electron pair. The electron's magnetic field is not clear and therefore have an atomic number of electrons, but it
```

Narrative continuation and topic drift example

```
Enter prompt: The rain continued through the night as the detective opened the old wooden door.

Generated Text:

The rain continued through the night as the detective opened the old wooden door. The Space Race spawned pioneering the
lunar surface, which was built in a dark-flight moment ago.  After Apollo 13, Apollo 13 landing gear and Apollo 15 had
landed in a row of land area, but it became known as the "the Moon". The Space Race (December 1972) on April 5, 1969. It was
built on July 6, 1969.  The Apollo 11 mission was launched by the crew of Apollo 13 and the backup crewed lunar landing site
in a day of the CSM, and had been flown at Kennedy Space Center (NASA), on April 7, 1972. The
```

Shakespeare-style generation behavior example

```
Enter prompt:  Why is the sky blue?

Generated Text:

 Why is the sky blue?     I'll tell thee, and be thy fair and thy youth; for thou art,    And make aught to be thyself. Thou
art thou so. Thou dost not; but the day of thy time, in my heart's womb!   KING HENRY. I will not speak no more: for the
hour of that time is a man, nor a fool.   KING HENRY. What art thou? Thou shalt be thyself, and thyself; and all thy
shalt not make thee an ass, but to
```

8. Experimental Analysis

The outputs indicate that the transformer successfully learned token relationships, sentence flow, and semantic structure. The model demonstrates behavior similar to early-stage GPT-style pretraining systems trained on diverse datasets.

The experiment also highlights the importance of: Dataset diversity Balanced domain sampling Efficient VRAM usage Stable attention mechanisms Sampling strategy tuning

9. Conclusion

eateggsAI-30M demonstrates that transformer-based language models can still be trained effectively on older consumer hardware with careful optimization and efficient pipeline design.

This project highlights the accessibility of AI research and shows that meaningful experimentation with transformer architectures is possible without enterprise-level hardware.

10. Future Work

Improved text generation quality Larger model experiments Inference optimization Fine-tuning support Chat interface integration Advanced attention optimization methods

Project Links

GitHub: https://github.com/eateggs0989/eateggsAI-30M
Hugging Face: https://huggingface.co/eateggs0989/eateggsAI-30M